

# Dhanush Pandiyan B

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## Education

MSc Biotechnology (2024 – 2026)

Sri Ramachandra Institute of Higher Education and Research, Chennai

CGPA: 8.61

BSc Botany (2021 – 2024) – Third Rank Holder

Thiagarajar College (Autonomous), Madurai

CGPA: 9.06

## Technical Skills

**Clinical Microbiology:** Microbial isolation, Pure culture techniques, Aseptic techniques, Media preparation, Gram staining, Microscopy, Culture maintenance, Microbial identification, Fermentation process optimisation, Laboratory-scale product development, Metabolomics.

**Molecular Biology:** PCR, DNA/RNA extraction, Agarose Gel Electrophoresis (AGE), RFLP analysis, primer designing

**Protein & Immunoassay Techniques:** ELISA, SDS-PAGE, Histostaining

**Laboratory Operations:** Laboratory documentation, QC record maintenance, Biosafety practices, contamination control, Experimental validation and data accuracy checks

**Cell Biology:** Mammalian cell culture handling, sterile cell culture techniques.

**Computational & Data Tools:** Julia programming (basic), Molecular docking, Machine learning, MS Office.

**Soft Skills:** Critical Thinking, Team Collaboration, Problem-Solving, Innovation Management

## Research Projects

### A. Microbiome research on tumor and oral wash from breast cancer patients

Investigating microbial virulence genes associated with breast cancer microbiome signatures.

Supervised by Dr Benedict Christopher Paul – SRIHER, Chennai, Tamil Nadu.

### B. Machine Learning for Viral Strain Prediction (Manuscript)

Developed a machine learning model using Julia to predict coronavirus strains based on spike protein physicochemical properties.

Supervised by Dr Benedict Christopher Paul – SRIHER, Chennai, Tamil Nadu.

### C. Instant Ferment Probiotic Product Development (Yukti)

Supervised by Dr M. Rama Prabha – TC, Madurai, Tamil Nadu.

## Experience

### A. National-level Summer Research Internship – Algal Biotechnology & Biodiversity taxonomy of algae and its application (BTAA23)

Madras Christian College (MCC), Chennai, May – Jul 2023

Conducted algal isolation, culture maintenance, and microscopy-based taxonomy analysis for industrial biotechnology applications.

### B. Research & Development Fellow – Fermentation Products

MakerGhat Foundation Incubation Program, Chennai, July – December 2024

Designed prototype instant fermentation-based probiotic products from laboratory to pilot scale.

### C. Research Assistant (ongoing)

CMRG, DoT, Chennai, started in August 2026 (3-year project); Department of Microbiology, Periyar University, Salem-11.

Entitled: Postbiotics from the kernels of *Mangifera indica* and *Syzygium cumini* waste from food processing industries alleviates type 2 diabetes mellitus

## **Achievements & Extracurriculars**

- Winner, TC Yukti 2024 Innovation Contest (Rs 5,000) – Recognised for product innovation.
- Academic Excellence Award, TC Madurai (Rs 12,500).
- Member, Anti-Ragging Council & NCC Naval Cadet (2021–2024).
- Volunteered for the university convocation at SRIHER.
- Presented an oral presentation – International Conference on Emerging Concepts in Biotechnology, SRM University.

## **Declaration**

I hereby declare that all information stated above is true and correct to the best of my knowledge and belief.

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